MILITARY SPECIFICATION SHEET

RECEPTACLE ASSEMBLY (MID-FLANGE, LOCKNUT, UNION TYPE)

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the receptacle assembly described herein shall consist of this document and the latest issue of MIL-C-24217.

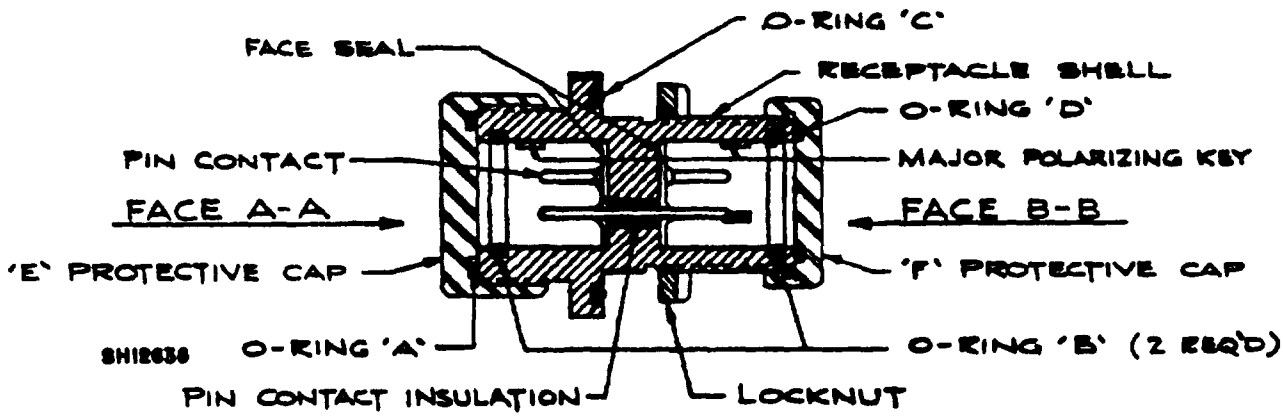


FIGURE 1. Assembly, receptacle.

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TABLE I. Parts description.

	RECEPTACLE ASSEMBLY NO.	RECEPTACLE SHELL NO.	NO. OF CONTACTS (FOR SIZE)	LOCKNUT NO. M24217/	O-RING SIZE AS-568A				"E" PROTECTIVE CAP PART NO.	"F" PROTECTIVE CAP PART NO.
					A	B	C	D		
	M24217/13-001	M24217/13-013	5 NO. 16	13-025	121	117	224	121	M24217/1-040	M24217/1-040
	M24217/13-002	M24217/13-016	14 NO. 16	13-026	129	125	229	129	M24217/1-042	M24217/1-042
	M24217/13-003	M24217/13-017	8 NO. 12	13-027	125	121	224	121	M24217/1-046	M24217/1-040
	M24217/13-004	M24217/13-018	3 NO. 0 (120° KEY)	13-028	146	141	238	146	M24217/13-040	M24217/13-040
	M24217/13-005	M24217/13-019	3 NO. 0 (150° KEY)	13-028	146	141	238	146	M24217/13-040	M24217/13-040

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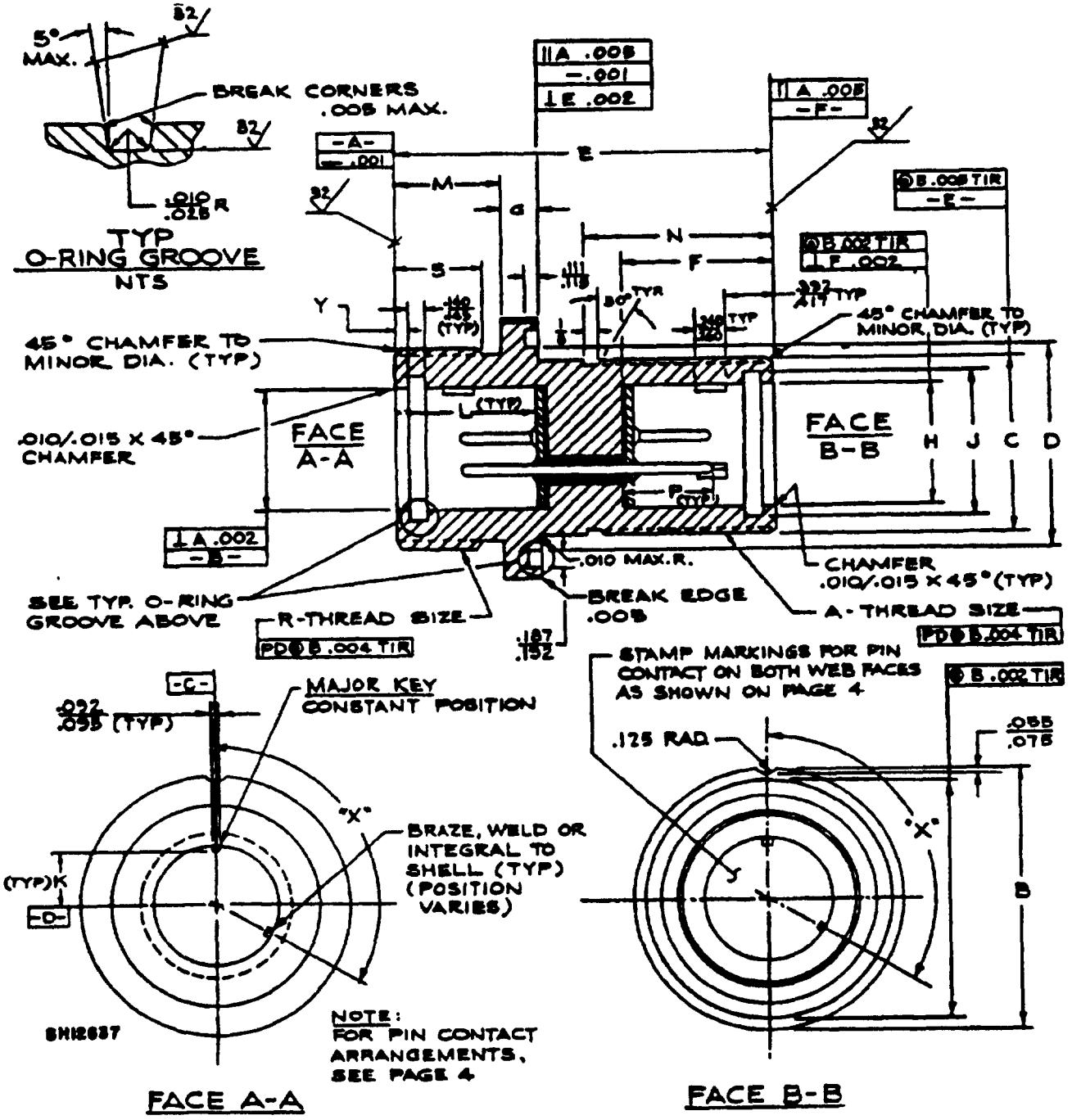


FIGURE 2. Receptacle shell (mid-flange, locknut, union type).

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TABLE II. Dimensions.

PART NO.	NUMBER OF CONTACTS & PIN SIZE	THD. SIZE R	THD. SIZE A	B	C	D	E	F	G	H	J	K	L	M	N	P	S	X	Y
M24217/13-015	5 No. 16	1 1/2-12 UN-2A	1 1/2-12 UN-2A	2.380 2.370	1.500 1.495	1.772 1.767	1.987 1.977	.797 .792	.312 .322	.818 .820	.984 .986	.312 .322	.740 .760	.370 .380	1.105 1.125	.313 .327	.270 .250	120°	.104 .109
M24217/13-016	14 No. 16	2 1/8-12 UN-2A	2 1/8-12 UN-2A	3.020 3.000	2.125 2.120	2.377 2.362	2.250 2.230	.797 .792	.315 .325	1.314 1.316	1.480 1.482	.574 .584	.720 .740	.490 .510	1.300 1.330	.313 .327	.270 .250	120°	.104 .109
M24217/13-017	3 No. 12	1 3/4-12 UN-2A	1 1/2-12 UN-2A	2.380 2.370	1.500 1.495	1.772 1.767	3.119 3.199	1.250 1.255	.315 .305	1.064 1.066	1.230 1.232	.448 .456	1.200 1.220	.870 .890	1.557 1.567	.750 .781	.707 .687	120°	.104 .109
M24217/13-018	3 No. 0	3 1/4-12 UN-2A	3 1/4-12 UN-2A	4.125 4.115	3.250 3.240	3.529 3.484	3.270 3.250	1.280 1.255	.326 .315	2.315 2.317	2.474 2.476	1.078 1.083	1.128 1.160	.930 .940	1.730 1.770	.780 .740	.750 .770	120°	.145 .150
M24217/13-019	3 No. 0	3 1/4-12 UN-2A	3 1/4-12 UN-2A	4.125 4.115	3.250 3.240	3.529 3.484	3.270 3.250	1.250 1.255	.325 .315	2.315 2.317	2.474 2.476	1.073 1.083	1.138 1.160	.930 .940	1.750 1.770	.750 .740	.750 .770	150°	.145 .150

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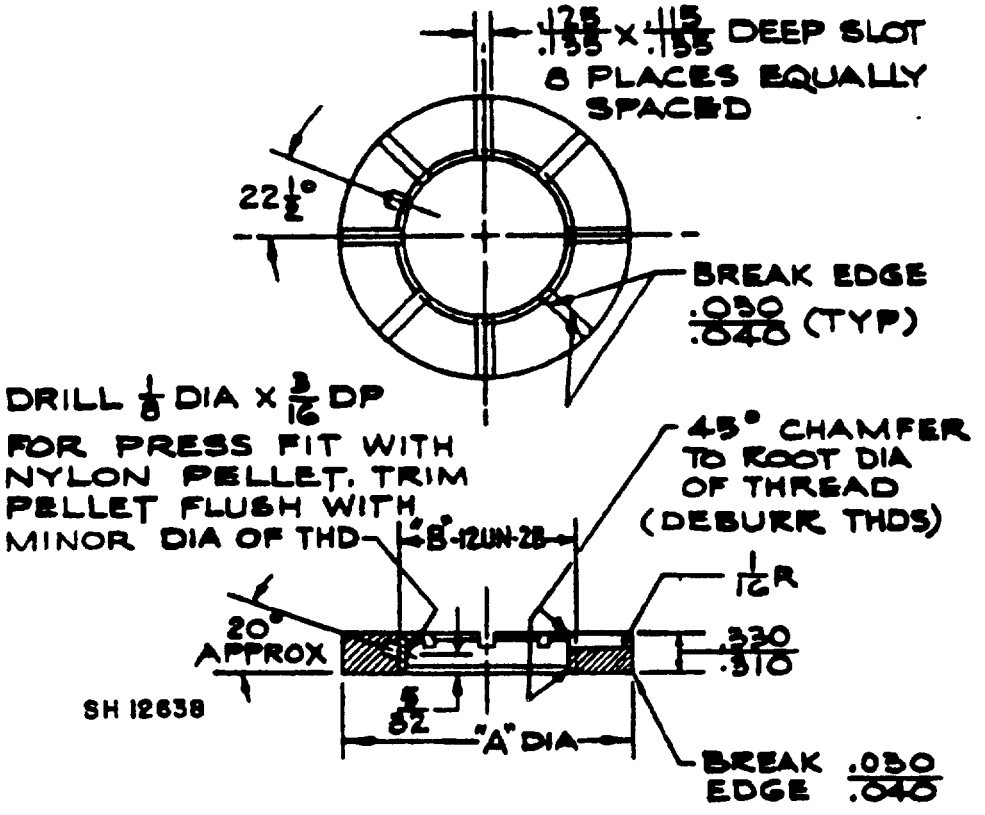


FIGURE 3. Locknut.

TABLE III. Dimensions.

PART NO.	"A" DIA	"B" THD
M24217/13-025	2.390/2.370	1 1/2
M24217/13-026	2.890/2.870	2 1/8
M24217/13-027	2.135/2.115	1 1/2
M24217/13-028	4.020-/4.000	3 1/4

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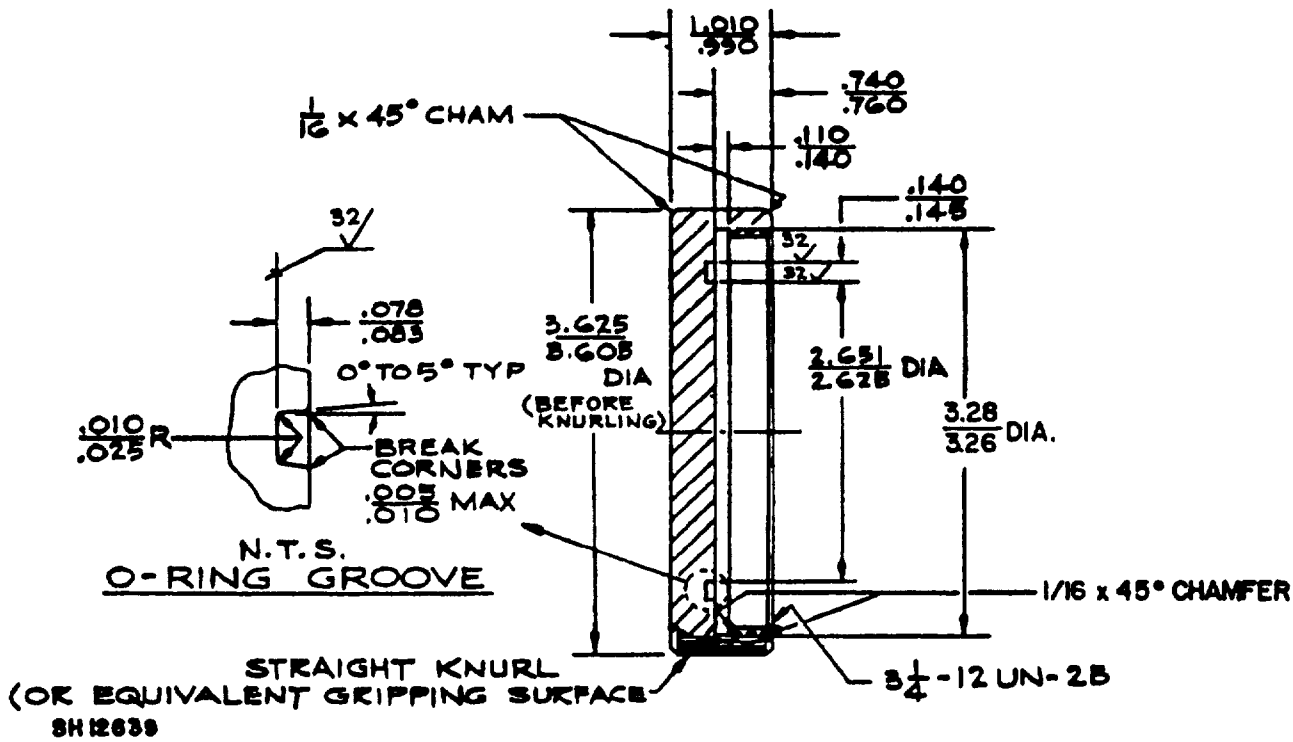
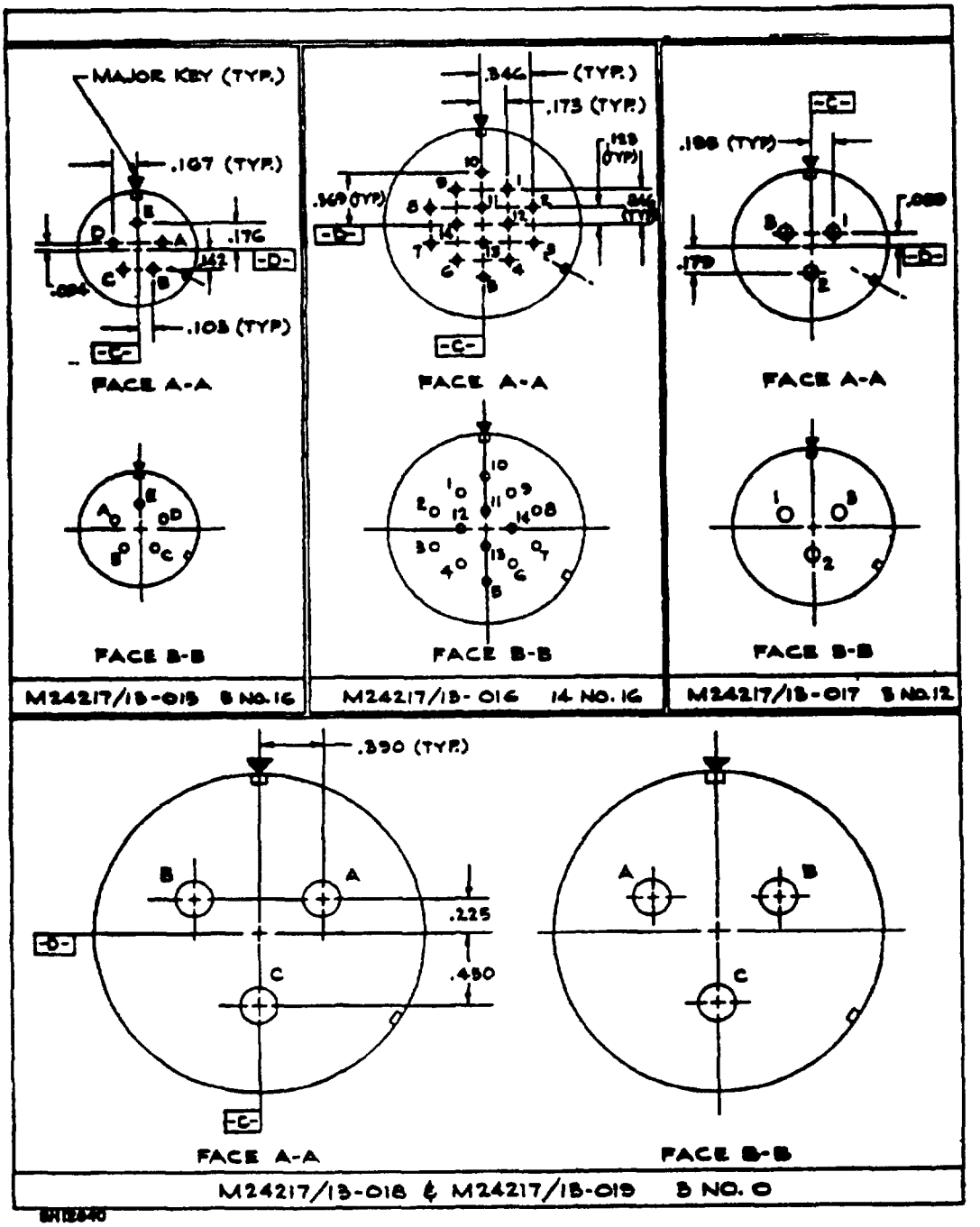


FIGURE 4. Protective cover, part number M24217/13-040.

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NOTES:

1. Datums -C- and -D- are vertical and horizontal centerlines of the receptacle bore diameter, their point of intersection having a true position within 0.003.
2. Pin contacts to be true position within 0.003 diameter to datums -C- and -D-.

FIGURE 5. Pin arrangements and markings.

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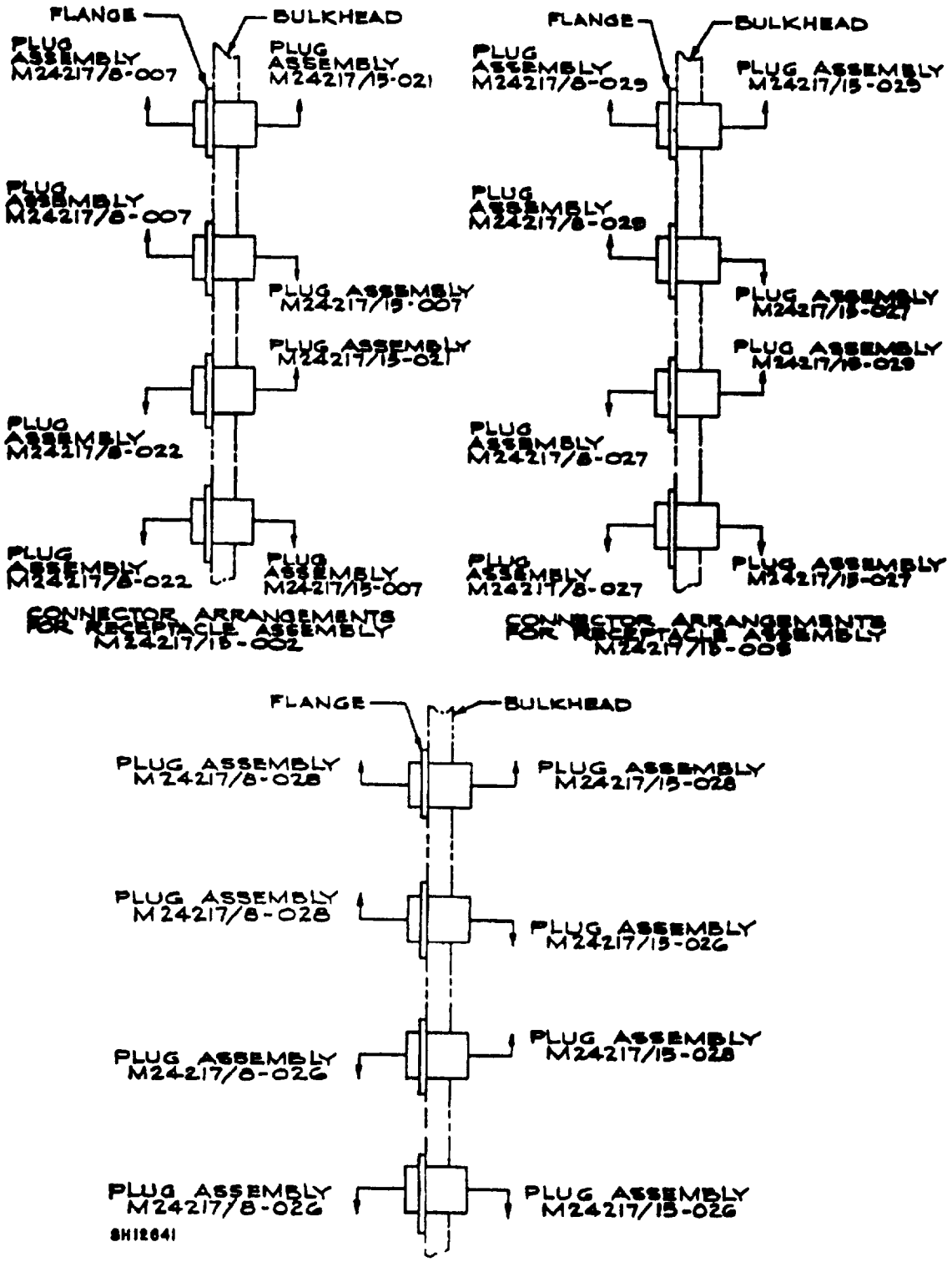
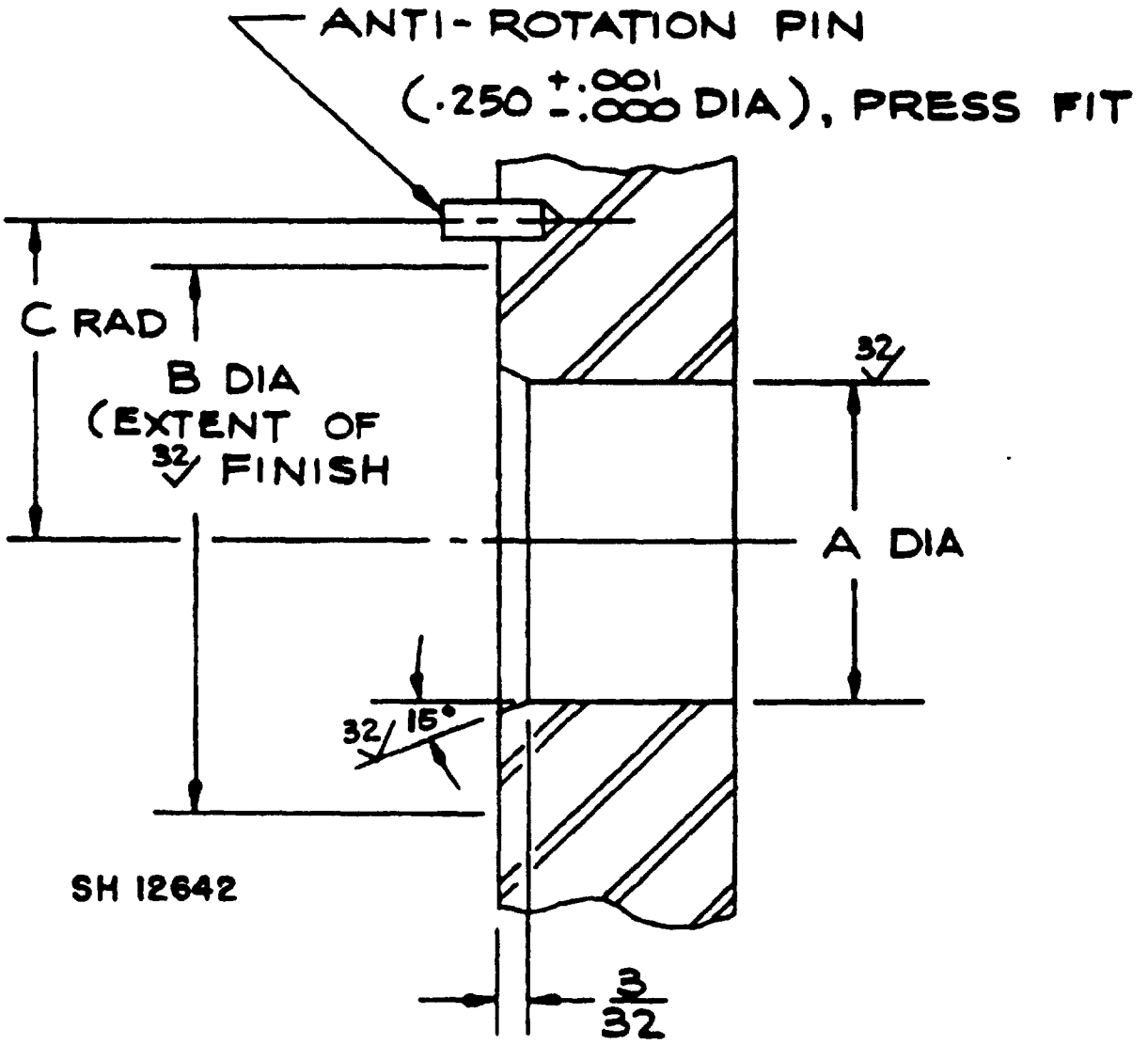


FIGURE 6. Connector arrangements for receptacle assemblies.

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NOTE: Bulkhead opening details are furnished for information purposes only. It is the responsibility of the installing activity to ensure that adequate hole compensation is provided in structural bulkheads, see requirement 2.

FIGURE 7. Bulkhead opening detail.

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TABLE IV. Bulkhead opening dimensions.

RECEPTACLE PART NO.	NO. OF CONTACTS & PIN SIZE	A DIA	B DIA	C RAD
M24217/13-015 M24217/13-017	5 NO. 16 3 NO. 12	1.515 1.520	2 1/4	1.260 1.265
M24217/13-016	14 NO. 16	2.140 2.145	2 7/8	1.580 1.583
M24217/13-018 M24217/13-019	3 NO. 0 3 NO. 0	3.265 3.270	4 1/16	2.132 2.135

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REQUIREMENTS:

1. Dimensions and configuration: See figures and tables.
2. Union receptacles are used as in-line, bulkhead, and other connector applications requiring a through, pin type penetration.
3. The material for the locknut shall be nickel aluminum bronze in accordance with QQ-C-465, alloy no. 630.
4. Contact gaskets (face seals) may be molded and secured to the receptacle web with adhesive or poured in place using a clear RTV silicone, General Electric RTV 615 with General Electric SS-4120 primer or equal. Either method shall prevent the entrance of water under the seal.
5. Locknut spanner wrenches shall be as shown on MIL-C-24217/6.
6. Receptacle assemblies shall be packaged as shown in the assembly view (protective covers tightened to compress the O-rings), except that the interior and flange O-rings shall be separately packaged.
5. Connector arrangements are shown on figure 6 of this document for receptacle assemblies having multiple right angle options with mating plug assemblies. For receptacle assemblies not shown, select the proper plug assembly from MIL-C-24217/7 (straight plugs) or MIL-C-24217/8 (right angle plugs) for the outboard or flange side and from MIL-C-24217/14 (straight union plugs) or MIL-C-24217/15 (right angle union plugs) for the inboard or locknut side. Table I of the appendix may be consulted for additional mating data.

Preparing activity:
Navy - SH
(Project 5935-N246-13)

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